

AI Meeting Action-Item Extractor

Python Project Report

A beginner-friendly application that converts meeting transcripts into structured action items.

1. Problem Statement

Meeting discussions often contain tasks, responsible people and deadlines in unstructured text. The purpose of this project is to automatically identify these action items and present them in a structured format.

2. Objectives

- Extract action-oriented statements from a transcript.
- Identify the task, owner and deadline.
- Assign a confidence value to each extracted item.
- Detect missing owners and deadlines.
- Remove duplicate tasks.
- Provide a simple upload/paste-to-results style interface.

3. Technologies Used

Python, Flask, HTML, CSS and regular-expression based NLP rules. The code is organized so a transformer/LLM layer can be added later.

4. System Workflow

Transcript → Cleaning → Sentence Segmentation → Action Detection → Owner/Deadline Extraction → Validation → Structured Results

5. Structured Output Schema

Field	Description
Task	Action that needs to be completed
Owner	Person responsible for the action
Deadline	Date or time mentioned for completion
Confidence	Estimated confidence of the extraction
Status	Open, Needs owner, or Needs deadline

6. Validation Rules

The program removes duplicate tasks, reports missing owners, reports missing deadlines, and flags invalid numeric date formats for checking.

7. Sample Input

Rahul will prepare the project report by Friday. Priya will review the report on Monday. John should send the client email by tomorrow.

8. Expected Result

The application displays each detected action item in a table with the task, owner, deadline, confidence and status.

9. How to Run

Install Python, open the project in VS Code, run **pip install -r requirements.txt**, then run **python app.py**. Open the local Flask address shown in the terminal.

10. Future Enhancements

A production version can add a real transformer/LLM model, speech-to-text for audio meetings, file upload for TXT/PDF/DOCX transcripts, database storage, user authentication, calendar integration and automated email reminders.

11. Conclusion

The project demonstrates how Python can transform unstructured meeting text into useful structured action items. Its modular design makes it suitable for beginner learning and future AI/NLP improvements.